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Autonomous Satellite Servicing Infrastructure for In-Space Assembly and Manufacturing

This work examines the closed-loop deployment of spacecraft servicing modules onto an uncontrolled bundle of assembly components as a method of gaining custody during the in-space assembly process. Minimum energy, low-thrust relative trajectories are generated using indirect optimal control on linear dynamics over a finite time search horizon. A line search solves for the minimum energy solution in the closed subset that satisfies a keep out zone constraint. A trajectory tracking linear quadratic regulator is used in conjunction with an extended Kalman filter to closed-loop maneuver the spacecraft under nonlinear relative orbital motion dynamics. Bearing angles, range, and range rate measurements are assumed available. Launch ΔV 's from the servicer spacecraft's deployment of a module and the ΔV 's seen in the relative frame upon impact on the client are included, and their effect on covariance due to impact site uncertainty is investigated. Discussions regarding these GN&C elements' importance within the context of the manufacturing/assembly process are included. [DOI: 10.1115/1.4067447]

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1 Introduction

Modern industrialized societies rely on large-scale production of goods to bring about economic progress and enhance quality of life for their citizens. Manufacturing, defined as the process of making articles and goods in large quantities, is at the heart of an industrialized society [1]. Design and manufacturing processes use repeatability, reliability, and resiliency as key performance criteria to evaluate the mechanization and automation options for effective production of articles in a variety of contexts [1,2]. Recent advances in digital engineering, robotics, and automation have proliferated a variety of paradigms, such as 3D printing, to bring unprecedented flexibility in manufacturing. In addition to creating innovative models where the low-volume production of bespoke articles becomes economically viable, these paradigms also enable substantial increase in product tolerance with minimal dilation of production cost. They also enable new manufacturing capabilities in extreme environments such as space manufacturing and assembly.

In-space assembly and manufacturing (ISAM) is an important capability that aims at exploiting the microgravity, low pressure, and vibration locale of the space environment to produce innovative applications [3,4]. Leveraging the affordability and accessibility of space enabled by emerging reusable launch technologies, ISAM focuses on two broad production categories. One category aims at

producing smaller scale articles that may be otherwise near-impossible or impractical to manufacture on Earth such as high-tolerance electronic components or sensitive bioengineering products. The other category aims at manufacturing and assembling mega-structures in space that would otherwise be too large to fit inside a launch vehicle payload fairing [5]. This category is particularly exciting because it represents a key enabling technology for future space exploration and space commercialization [6]. Manufacturing either category in the challenging environment of space typically entails parallel advances in space operations and technologies to support the salient aspects of manufacturability. Repeatability of a process involves the ability of the component technologies to support a sustained production of multiple articles from raw or semi-finished materials in the relevant environment. In traditional manufacturing applications, repeatability of a process enables the production of identical articles in a larger scale to support a scale-up of the production process [7]. For in-space manufacturing, the use of digital models and high-speed data links will create paradigm shifts in the notion of repeatability by enabling component production in the remote space environment. These facilities will eventually be more economical to produce at the point of need because the launch, transit, and delivery costs are minimized to deliver the product at the required location. Similarly, resiliency and reliability are important elements of a manufacturing process since one desires the process to be robust to changes in the supply of the semi-finished or raw materials in the production line [8]. This is an especially important consideration for space applications where the replenishment of the manufacturing unit may

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depend on resupply missions. Fundamentally, robustness and repeatability of a space manufacturing process cannot be achieved without advances in the component technologies of robotics and automation.

This article focuses on the ISAM of mega-structures. For autonomous relative maneuverability in space, a vehicle must be fitted with a variety of subsystems including power and propulsion units, rendezvous/proximity operations/docking (RPOD) sensor suites, attitude determination and control, and communication links. It is economically impractical to fit these types of subsystems to every structural component or bundle of components launched into space to construct a particular product. Rather, it is more appropriate to design these bundles with capture points for a servicing spacecraft or swarm of servicing spacecraft to grab hold of in order to autonomously carry out the assembly process. This problem falls under the broad category of gaining custody of *uncontrolled* resident space objects (RSOs). Methods to solve this problem often involve the use of robotic manipulators fixed to servicing spacecraft because they can be used to carry out servicing/manufacturing activities after custody is achieved. Methods assuming the RSO's geometry and mass distribution are well understood and include optimal joint control of a serial manipulator post-hard capture [9], or constant contact control pre-hard capture in which end-effector joints acting as spring-mass-damper systems dissipate the target's rotational kinetic energy [10]. RSO mass distribution may be uncertain even if the geometry is known which can cause control instability or violate force/torque limits. To alleviate this issue, a robust compensator can be designed around the inertial parameter uncertainty bounds to guarantee system stability [11]. Alternatively, an adaptive control system can be applied to simultaneously estimate inertia parameters while performing spin stabilization with constant contact [12]. These methods provide quick means of gaining custody, but assume that the RSO's angular velocities are low enough such that the manipulator or its spacecraft bus can track the motion of the desired contact or capture point. In the general sense for an uncontrolled target, this cannot be assumed, and the use of manipulator methods centered on hard contact risk damage or debris events. This lowers a system's resilience to unexpected mission criteria such as the loss of an RSO's reaction wheel or control moment gyro. Moreover, a singular servicing spacecraft with a robotic manipulator(s) is limited to dealing with one RSO at a time. From an ISAM perspective, this is inefficient.

Contactless approaches to detumbling uncontrolled RSOs come in a wide variety. Active electrical charge transfer between servicer and RSO can be directed so as to gradually apply spin stabilizing control torques, though the time scale is tens of days [13]. Another method employs a capture net and elastic tether to stabilize a tumbling body [14], though the dynamical interaction between these systems is poorly understood, and lends itself specifically to space debris. Lastly, the European Space Agency studied the use of a servicer's thruster out-gas to impart force-torque pairs against the angular velocity of the RSO, with an experiment making it through its proposal stage (although it was canceled because of diminishing propellant budget used during satellite's primary mission) [15].

A novel approach to the uncontrolled, servicing/assembly problem that has recently been investigated involves the deployment of RPOD-oriented, thrust capable modules from a host spacecraft [16,17]. Modules are launched from the servicer spacecraft toward a desired point of impact on an RSO where they then attach. Each module is equipped with a simplified RPOD sensor suite that allows for terminal navigation and navigation between modules around the RSO [18]. Impacts and attachments alone can act to slow the RSO's angular body rates through conservation of angular momentum. However, if body rates are high, modules can thrust in moment-couple pairs using cubeSat level maneuver packs to further null out a RSO's angular momentum while estimating its inertial parameters [19]. Safe attachment methods are an active area of research, but the use of gecko inspired materials [20] or active electromagnetics has been proposed. Both attachment methods allow for the disconnection of modules after the RSO has

been stabilized and/or assembled. For tumbling RSOs, this method of gaining custody decouples servicer hardware from the act of detumbling, lowering risk associated with the more expensive system of systems. The responsibility for this sub-task is given to the smaller, more maneuverable modules, whose individual costs are lower. In addition, because modules are being launched toward the RSO, prediction of its attitude motion can be used to line up a launch/impact in the desired area such that the servicer need not track the RSO's attitude with relative orbital motion. Because of the thrust capability of the modules once attached, this approach to gaining custody can be completed in a time sensitive manner. Lastly, this servicing/assembly architecture supports the handling of multiple RSOs simultaneously, leading to a resilient and efficient manufacturing procedure. One limitation of this approach lies in the existence of a suitable RSO impact/attachment site. While the attachment method can be compliant, a surface rigidly connected to the RSO must be present and identifiable with enough surface area acceptable to handle position uncertainties upon impact. If the RSO is simply a bundle of raw materials, a large monolithic spacecraft, or debris like a spent rocket body, these impact sites are likely available.

Drawing parallels and analogies with the mechanical devices of the early industrial revolution, control systems technology evolved to mechanize the operation of early production equipment [21]. While Watt's original steam governor was a mechanical system, subsequent advances led to manufacturing automation controlled by electrical and electronic control loops [22,23]. To this end, the article presents prototype space manufacturing approaches that leverage simultaneous advances in microelectronics, sensors, and actuators. Mature digital avionics platforms and wireless communication devices are used to demonstrate automated operation of the spacecraft modules to perform key manufacturing functions. The technical demonstrations of the article that utilize the prototype space automation units form a strong basis for optimism in the deployment of such platforms for in-space assembly and manufacturing applications. Approaches to evaluate the performance of the operational prototypes will be utilized to study their utility and applicability in advanced space manufacturing applications.

1.1 Contribution. The main contribution of this work is the introduction of the deployed module approach to the uncontrolled, in-space servicing, assembly and manufacturing problem. Specifically, a high-level, closed-loop guidance, navigation, and control synthesis of several previously developed subsystems [16–19] is presented in order to demonstrate the concept of operations involved in this distributed system approach.

2 Scope and Limitations

This article covers the guidance, navigation, and control activity required to deploy modules to predefined locations on a tumbling RSO which represents a bundle of components to be used in the assembly of an orbiting mega-structure. A high-level concept of operations is outlined in Fig. 1, initialized by a near-rendezvous mission phase. The module impact sites and their surface normal vectors are assumed to be computed in the planning phase of this mission prior to deployment. Open-loop guidance using a minimum energy indirect optimal control formulation with linearized equations of relative motion is wrapped into a line search algorithm bounded by a defined launch cadence to compute energy optimal paths that obey keep out zone constraints. An extended Kalman filter (EKF) and a multiplicative extended Kalman filter (MEKF) are used with relative right ascension and declination angles, range/range rate measurements, and relative quaternion measurements to estimate the relative spacecraft state with respect to the RSO's local vertical local horizontal (LVLH) frame. All decision-making algorithms within Fig. 1 utilize these estimated states, which are by nature uncertain. Lastly, closed-loop, linear quadratic regulator (LQR) model reference control is employed to

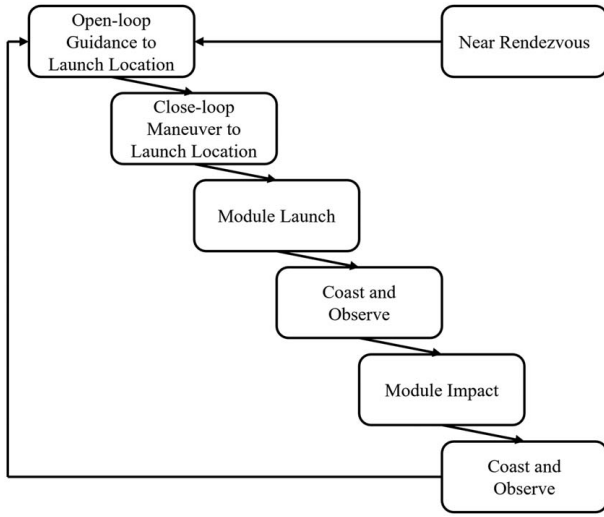


Fig. 1 GN&C concept of operations for deployment of satellite servicing modules

track the open-loop path under nonlinear dynamics given the uncertain state estimates from the EKF, and such a control scheme is shown to be robust to these state uncertainties. Effects of module launches and module impacts on the relative motion seen by the servicer are discussed. Results from a closed-loop simulation for a tumbling bundle of materials with the size/mass of a global position system (GPS) satellite in mid-Earth orbit are shown.

3 System Model

Interacting with a RSO in space involves the attitude and orbital motion of the servicer relative to the RSO as they both orbit a primary body (i.e., Earth). Rather than viewing this relative motion in an Earth centered inertial frame $\mathcal{N}$, it is more convenient both mathematically and intuitively to look at relative motion in the LVLH frame $\mathcal{L}$ of one of the two spacecraft involved—typically the passive spacecraft in the scenario (RSO in this case). The LVLH frame has $\hat{x}$ component in the RSO's radial direction, $\hat{z}$ component in the direction of the RSO's orbital angular momentum, and $\hat{y}$ component completing the right-handed triad pointing loosely in the direction of orbital velocity. This is shown in Fig. 2, where $\mathbf{R}$ is the position of the RSO relative to Earth center, and $\mathbf{r} = [x \ y \ z]^T$ is the position of the servicer relative to the RSO.

3.1 Relative Orbital and Attitude Motion. Assuming only two-body (Keplerian) accelerations under no orbital perturbations, the full, nonlinear equations of relative orbital motion for the servicer with respect to the RSO expressed in the RSO's LVLH frame are used for simulation in this work and can be found in Ref. [24]. The RSO's orbital radius R and velocity projected in the radial direction $\dot{R}$ must be propagated alongside the servicer's relative orbital states in this model. The servicer's control vector

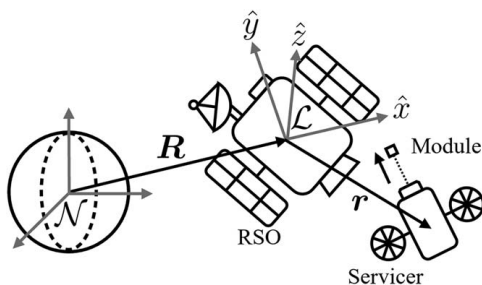


Fig. 2 Relative orbital motion model

$\mathbf{u} = [u_x \ u_y \ u_z]^T$ takes the form of continuous accelerations expressed in the RSO's LVLH frame. The resulting system of coupled, nonlinear differential equations contains eight states $\mathbf{x} = [x \ y \ z \ \dot{x} \ \dot{y} \ \dot{z} \ R \ \dot{R}]^T$, rather than 12 if relative motion was examined solely in an inertial frame.

Under two fundamental assumptions, the nonlinear equations of relative motion can be reduced to a linear system known as the Clohessy–Wiltshire equations [25]. The first assumption claims that the RSO vehicle resides in a circular orbit with eccentricity $e = 0$. The second assumption claims that the servicer's relative distance is several orders of magnitude smaller than the orbital radius of the RSO. These assumptions result in Eq. (1), with n as the RSO's mean orbital motion, and the relative state vector $\mathbf{x} = [x \ y \ z \ \dot{x} \ \dot{y} \ \dot{z}]^T$.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{0}_{3 \times 3} \\ \mathbf{I}_{3 \times 3} \end{bmatrix} \quad (1)$$

Equally important as relative orbital motion is the relative attitude motion between the servicer and RSO spacecraft. For this work, the servicer is treated as a rigid body whose module deployment mechanism is aligned with its mass center such that no attitude disturbance is created when modules are launched. In tracking the rigid body attitude of the RSO, relative pose measurements are assumed to be transformed into an assigned RSO's body fixed frame $\mathcal{B}$ with respect to the inertial frame $\mathcal{N}$ before being processed for use in estimation and control. Because of this, only the passive RSO's attitude motion is considered explicitly. Attitude control of the servicer is assumed exact and not included in this analysis. See Ref. [26] for more on attitude dynamics.

The RSO's inertia tensor $\mathbf{I}^{\mathcal{B}}$ is assumed known to start (cooperative scenario), though it can be learned through system excitation as modules are deployed [16,19]. The quaternion is chosen as a non-minimal, globally non-singular attitude representation describing the orientation of $\mathcal{B}$ with respect to $\mathcal{N}$.

$$\bar{\mathbf{q}}_{\mathcal{N}}^{\mathcal{B}} = \begin{bmatrix} \mathbf{q} \\ q \end{bmatrix} \quad (2)$$

The kinematics associated with the quaternion formulation appear in Eq. (3):

$$\dot{\bar{\mathbf{q}}}_{\mathcal{N}}^{\mathcal{B}} = \frac{1}{2} \begin{bmatrix} \boldsymbol{\omega}^{\mathcal{B}} \\ 0 \end{bmatrix} \otimes \bar{\mathbf{q}}_{\mathcal{N}}^{\mathcal{B}} \quad (3)$$

where the operator $\otimes$ represents quaternion multiplication, and $\boldsymbol{\omega}^{\mathcal{B}}$ is the RSO's angular velocity in its body frame.

3.2 Frame Transformations. All relative motion, both attitude and orbital, is viewed in the RSO's LVLH frame. To convert between $\mathcal{B}$, $\mathcal{N}$, and $\mathcal{L}$, several frame transformations are defined. To transform vectors from $\mathcal{N}$ to $\mathcal{B}$, a rotation matrix in terms of the unit quaternion components is used.

$$\mathbf{R}_{\mathcal{N}}^{\mathcal{B}} = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1q_2 - q_3q_4) & 2(q_1q_3 + q_2q_4) \\ 2(q_1q_2 + q_3q_4) & 1 - 2(q_1^2 + q_3^2) & 2(q_2q_3 - q_1q_4) \\ 2(q_1q_3 - q_2q_4) & 2(q_2q_3 + q_1q_4) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix} \quad (4)$$

The frame transformation from $\mathcal{N}$ to $\mathcal{L}$ is a function of the RSO orbit's right ascension of the ascending node Ω , inclination i , argument of periape ν , and true anomaly f . This transformation is then defined by the following 3-1-3 Euler angle sequence.

$$\mathbf{R}_{\mathcal{N}}^{\mathcal{L}} = \mathbf{R}_3(\nu + f) \cdot \mathbf{R}_1(i) \cdot \mathbf{R}_3(\Omega) \quad (5)$$

Finally, noting that for a proper orthonormal rotation matrix $R^{-1} = R^T$, the transformation from $\mathcal{B}$ to $\mathcal{L}$ is written as Eq. (6).

$$\mathbf{R}_B^{\mathcal{L}} = \mathbf{R}_N^{\mathcal{L}} \cdot [\mathbf{R}_N^{\mathcal{B}}]^T \quad (6)$$

4 Guidance

The manufacturing, assembly, and servicing of components in space are ultimately limited by the maneuvering capability required to hard-dock two or more free floating bodies. This takes the form of either a finite supply of fuel/propellant or time-to-charge in the case of solar-electric propulsion. Optimal spacecraft trajectories must be computed that minimize performance objectives related to these quantities in order to maximize product or service capability. Many methods for spacecraft relative motion path planning have been proposed [27–30]. The method employed in this article leverages the computational efficiency of indirect optimal control formulations of linear dynamical systems to compute and refine locally fuel optimal trajectories that meet mission specified time constraints [31,32]. While solution existence is not guaranteed under this method, it has been shown that with the proper choice of system parameters (keep out zone radius, launch distance, and nominal launch cadence), solutions readily exist [16] and the solution space is robust under realistic operating parameters/regimes.

4.1 Indirect Optimal Control. Using an indirect optimal control formulation [33], a trajectory optimization problem is converted into a two point boundary value problem (TPBVP) subject to constraints. The result takes the form of an initial set of “costates” λ , augmented states that relate to the control input, that create the open-loop guidance solution to the TPBVP. For the optimization of spacecraft trajectories, cost functions over the trajectory usually contain some term related to the control effort because fuel/propellant limits the lifespan of the vehicle. To maneuver between deployment states in the RSO’s relative frame, a minimum energy cost function is constructed:

$$J = \int_{t_0}^{t_f} \frac{1}{2} \mathbf{u}^T \mathbf{u} \, dt = \int_{t_0}^{t_f} L(\mathbf{u}) \, dt \quad (7)$$

To minimize this cost function with respect to a set of governing dynamics $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ and terminal boundary constraints $\psi(\mathbf{x}(t_f))$, the augmented cost function is defined:

$$J_a = \psi(\mathbf{x}_f) + \int_{t_0}^{t_f} [L(\mathbf{u}) + \lambda^T (\mathbf{f}(\mathbf{x}, \mathbf{u}) - \dot{\mathbf{x}})] \, dt \quad (8)$$

Taking the first variation of parameters with a fixed final time and setting arbitrary variations to zero gives the necessary conditions (Eqs. (9) and (10)), which when solved guarantee the solution is an extrema to Eq. (7).

$$\dot{\lambda} = -\left(\frac{\partial L}{\partial \mathbf{x}} + \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \lambda\right) \quad (9)$$

$$\frac{\partial L}{\partial \mathbf{u}} = -\frac{\partial \mathbf{f}}{\partial \mathbf{u}}^T \lambda \quad (10)$$

The TPBVP is then stated as: given t_0 , t_f , and $\mathbf{x}(t_0)$, solve for $\lambda(t_0)$ such that $\psi(\mathbf{x}(t_f)) = 0$, where ψ is generally a vector valued nonlinear function of the terminal states. This TPBVP can be solved using a differential corrector based on a first-order Newton’s method either by numerical shooting or collocation—provided a sufficient initial guess is available.

Thus far, no specific dynamics have been stated. The nonlinear equations of relative motion could be used, and the resulting trajectory would obey the true two body dynamics involved. However, with non-zero eccentricity, seeding this nonlinear TPBVP with a sufficient initial guess for algorithmic convergence is nontrivial. Moreover, computation time is strongly dependent on the initial guess provided. And this is with a fixed time of flight. Converting

this to a free final time problem would only exacerbate these issues. Using the linear relative equations of motion of Eq. (1), these computational difficulties are alleviated, as the solution is shown to reduce to simply a matrix inversion if integral pseudospectral collocation is used [31]. Unfortunately, this comes at a higher control cost when the loop is closed with feedback control because the desired trajectory provided by linear open-loop guidance does not obey the nonlinear vector field. Since optimal linear trajectories can be computed through a single matrix inversion, a globally near-optimal nonlinear trajectory can be rapidly recomputed on the fly such that at the local level, each segment is well approximated by its linear counterpart [32]. Using Eq. (1), the necessary conditions reduce to Eqs. (11) and (12)

$$\dot{\lambda} = -A^T \lambda \quad (11)$$

$$\mathbf{u} = -B^T \lambda \quad (12)$$

This fixed time linear formulation is then nested inside a line search over a finite time horizon to find the minimum energy linear trajectory that obeys a keep out zone constraint.

4.2 Open-Loop Module Deployment: Initialization. The goal of this guidance algorithm is to find the minimum energy path that meets both time and space constraints. This algorithm assumes that a particular mission requires a nominal module launch cadence t_c . This could be a function of mission importance, lighting conditions, Earth communication availability, or control authority. With a maximum deviation time bounding the cadence t_b , the time at which launch can occur must then lie in the finite set $t_l \in [t_c - t_b, t_c + t_b]$. To initiate the algorithm, this set is split evenly into N possible launch times. For each launch time, the minimum energy, linear TPBVP is constructed using estimated initial relative orbital states and RSO attitude states. The resulting trajectory cost from Eq. (7) is computed using quadrature. To ensure safe RPOD activity, intersection of the trajectory with a specified keep out sphere with radius r_k is also verified such that:

$$x^2 + y^2 + z^2 > r_k^2 \quad (13)$$

If the spatial constraint is met, the cost is compared to the previous solutions and saved if smaller.

Prior to any module deployment, it is assumed that RSO impact sites $\mathbf{r}_i^{\mathcal{B}}$ with corresponding surface normal vectors $\hat{\mathbf{n}}_i^{\mathcal{B}}$ have been selected to suit a certain ISAM mission. Additionally, a launch distance r_l away from every impact site is specified for safety and observation, along with the launch velocity magnitude v_l of the module deployed from the servicer. The launch velocity magnitude is assumed to be several orders of magnitude larger than the relative velocity magnitude of the servicer. Then, for the time scale difference between orbital motion and module time aloft, it is appropriate to assume that module trajectories follow straight line paths in the RSO’s LVLH frame. This then fixes the module’s time aloft to be $t_m = r_l/v_l$.

To construct each TPBVP, the RSO’s attitude is propagated forwards to the impact time

$$t_i = t_l + t_m \quad (14)$$

using the estimated attitude states at the current time. The relative launch position for the servicer is computed in the RSO’s body frame as

$$\mathbf{r}_l^{\mathcal{B}} = \mathbf{r}_i^{\mathcal{B}} + r_l \cdot \hat{\mathbf{n}}_i^{\mathcal{B}} \quad (15)$$

and then transformed into the RSO’s LVLH frame using the predicted RSO quaternion at impact time and the transformation in Eq. (6).

$$\mathbf{r}_l^{\mathcal{L}} = \mathbf{R}(\bar{\mathbf{q}}_B^{\mathcal{L}}(t_i)) \cdot \mathbf{r}_l^{\mathcal{B}} \quad (16)$$

This geometry is illustrated in Fig. 3.

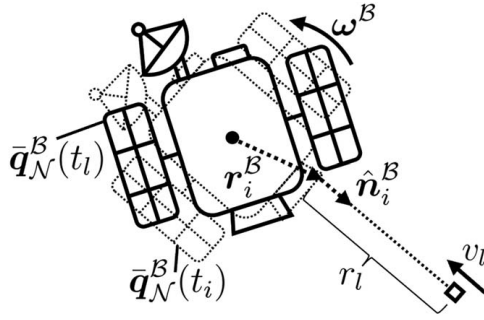


Fig. 3 Launch geometry

With the launch position fixed for each launch time in the set, a corresponding launch velocity must be chosen to complete the state, and hence the TPBVP. The bounded linear relative motion condition is used to constrain velocity in the in-track direction. Forcing the resulting relative motion ellipse to be centered at the RSO constrains velocity in the radial direction [24]. These selections permit safe launch aborts: if the servicer does not launch a module at the launch time and drifts under natural motion, its trajectory will stay close to the RSO for some local time constant until nonlinear effects cause divergence. Lastly, out-of-plane velocity is set to zero at launch to reduce control requirements post launch if the servicer needs to return to a nominal inspection orbit.

$$v_l^c = \left[\frac{1}{2}ny_l, -2nx_l, 0 \right]^T \quad (17)$$

Another choice for servicer launch velocity selection would be to match the linear velocity of the impact site on the RSO due to its angular velocity so that the transverse relative velocities between the module and RSO body are minimized upon impact. This, however, would be highly dependent on the servicer's estimate of the RSO's attitude motion, which is imperfect at best, even with exact knowledge of RSO inertial parameters. Furthermore, if the RSO was tumbling with large body rates, the relative orbital motion control authority required to match this may be unachievable. Instead, it is assumed that the terminal guidance system in the modules themselves mediate the non-zero transverse relative velocity upon impact, and that the attachment method is robust to this unavoidable circumstance. The guidance initialization algorithm is summarized as follows:

Algorithm 1 Guidance solution: initialization

```

1: Initialize cost  $J^*$  and solution  $\mathcal{G}^*$ 
2: for  $t_l \in [t_c - t_b, t_c + t_b]$  do
3:   Propagate target attitude to time  $t_l$ 
4:   Compute  $R_B^c$  using Eq. (6) at time  $t_l$ 
5:   Compute servicer launch state using Eqs. (15), (16), and (17) associated with the current impact site
6:   Obtain TPBVP minimum energy optimal guidance  $\mathcal{G}$  using Eqs. (11) and (12)
7:   if any  $\|r^c(t)\| \in \mathcal{G} < r_k$  then
8:     Keep out zone violated. Discard trajectory
9:    $t_l = t_l + \frac{2t_b}{N}$ 
10:  else if  $J < J^*$  then
11:     $\mathcal{G}^* = \mathcal{G}$ ,  $J^* = J$ 
12:   $t_l = t_l + \frac{2t_b}{N}$ 
13:  end if
14: end for

```

The output to Algorithm 1 is then a particular set of initial conditions for the costates in the indirect optimal control formulation along with the associated launch time that is energy optimal over

the discretized grid while obeying the keep out zone constraint. Note that solution existence is not guaranteed. However, with proper parameter selection, i.e., making the launch distance sufficiently larger than the keep out radius as well as a large enough nominal launch time, the method is generally robust as the minimum energy solution leans more and more on natural motion with increased time of flight [16]. An example of the output for this initialization stage for $N = 50$ is shown in Fig. 4 where trajectories in pink violate the keep out zone, trajectories in black do not, and the gray sphere is the keep out zone itself.

4.3 Open-Loop Module Deployment: Refinement. From the output grid of trajectories, Algorithm 1 returns the discrete optimal path $\mathcal{G}^*$. To obtain a locally optimal solution over the continuous time domain, this solution is refined using a one parameter line search via method of bisections. The two neighboring grid points t^- and t^+ surrounding the discretized solution time t^* are used to initialize the line search bounds, assuming they themselves were viable trajectories. With a fine enough initial grid and acknowledging continuity of the system's dynamics (both RSO attitude and servicer relative motion), trajectories inside the search bounds are not only viable themselves but also are guaranteed to include a local optimal solution. The refinement algorithm is then summarized in Algorithm 2.

Algorithm 2 Guidance solution: refinement

```

1: Given  $t^-, t^+, \Delta t_{tot}$ 
2: while  $\Delta t > \Delta t_{tol}$  do
3:    $\hat{t} = \frac{1}{2}(t^- + t^+)$ ,  $\Delta t = \frac{1}{2}(t^- - t^+)$ 
4:   Compute  $\mathcal{G}^-, \mathcal{G}^+$ , and  $\mathcal{G}$  and corresponding costs with the subroutine in Algorithm 1.
5:    $\Delta J^+ = |J^+ - \hat{J}|$  and  $\Delta J^- = |J^- - \hat{J}|$ 
6:   if  $\Delta J^- > \Delta J^+$  then
7:      $t^- = \hat{t}$ 
8:   else
9:      $t^+ = \hat{t}$ 
10:  end if
11: end while

```

With the optimal open-loop trajectory selected, state estimates from the relative navigation filters can be used along with a feedback controller to maneuver the servicer to the determined launch location.

5 Navigation

In order to achieve a closed-loop, resilient system architecture that is reliable under system uncertainties, it is vital that relevant states of the RSO and servicer are accurately estimated. This section details the estimation methods utilized, as well as the relevant measurements and associated assumptions.

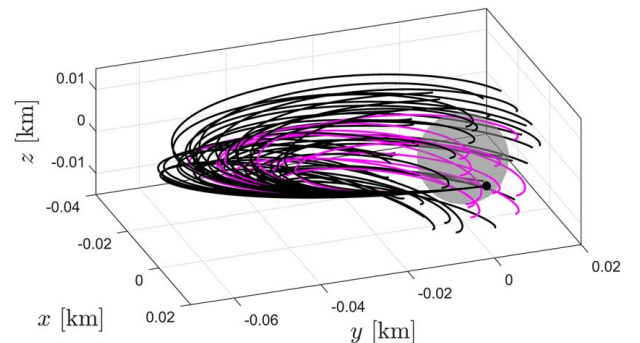


Fig. 4 Example output of guidance solution initialization

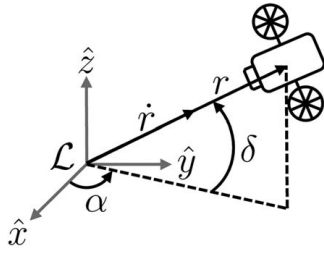


Fig. 5 Right ascension, declination, and range and range rate measurement geometry for relative orbital motion EKF

5.1 Measurement Models

5.1.1 Right Ascension and Declination Angles. One method to define angle measurements for relative navigation or orbit determination is by using two observation angles, namely the right ascension and declination angles, which are represented by α and δ , respectively. These angles are typically backed out of pixel locations in images taken from monocular or stereoscopic, visible or infrared cameras. Figure 5 plots the geometry of this configuration. The measurement function takes the form

$$\mathbf{h}_{ba}(\mathbf{x}) = \begin{bmatrix} \alpha \\ \delta \end{bmatrix} = \begin{bmatrix} \tan^{-1}\left(\frac{y}{x}\right) \\ \tan^{-1}\left(\frac{z}{\sqrt{x^2 + y^2}}\right) \end{bmatrix} \quad (18)$$

The observation angle measurements are represented as

$$\mathbf{z}_{ba,k} = \mathbf{h}_{ba}(\mathbf{x}_k) + \mathbf{v}_{ba,k} \quad (19)$$

where $\mathbf{v}_{ba,k}$ is the noise associated with the observation angle measurements, and is taken from a zero mean Gaussian random distribution with covariance

$$\Sigma_{ba,v} = \sigma_{ba,v}^2 \mathbb{I}_{2 \times 2} \quad (20)$$

Based on Ref. [34], the standard deviation of the measurement noise, $\sigma_{ba,v}$, is 0.03 deg. Measurements are received at a rate of 0.1 Hz.

5.1.2 Range. Velocimeter light detection and ranging (LIDAR) sensors are capable of providing range and range rate measurements via point cloud and Doppler velocity measurements. These sensors are commonly found in a fully outfitted RPOD sensor suite. The measurement function for range is given by

$$h_r(\mathbf{x}) = r = \sqrt{x^2 + y^2 + z^2} \quad (21)$$

The range measurements are represented as

$$z_{r,k} = h_r(\mathbf{x}_k) + v_{r,k} \quad (22)$$

where $v_{r,k}$ is the noise associated with the range measurements, and is taken from a zero mean Gaussian random distribution with covariance

$$\Sigma_{r,v} = \sigma_{r,v}^2 \quad (23)$$

Based on Ref. [35], the standard deviation of the measurement noise, $\sigma_{r,v}$, is 0.05 m. Measurements are received at a rate of 10 Hz but are downsampled to a consistent rate of 0.1 Hz.

5.1.3 Range Rate. Taking the time derivative of the range, range rate is obtained as well, whose measurement function is given by

$$h_{rr}(\mathbf{x}) = \dot{r} = \frac{x\dot{x} + y\dot{y} + z\dot{z}}{\sqrt{x^2 + y^2 + z^2}} = \frac{x\dot{x} + y\dot{y} + z\dot{z}}{r} \quad (24)$$

Thus, range rate measurements can be represented as

$$z_{rr,k} = h_{rr}(\mathbf{x}_k) + v_{rr,k} \quad (25)$$

where $v_{rr,k}$ is the noise associated with the range measurements, and is taken from a zero mean Gaussian random distribution with covariance

$$\Sigma_{rr,v} = \sigma_{rr,v}^2 \quad (26)$$

Based on Ref. [35], the standard deviation of the measurement noise, $\sigma_{rr,v}$, is 0.1 m s^{-1} . Again, measurements are received at a rate of 10 Hz but are downsampled to a consistent rate of 0.1 Hz.

5.1.4 Relative Attitude. For spacecraft proximity operations, one of the state-of-the-art methods for relative pose estimation is through the use of computer vision (CV) via neural networks. Implementing such algorithms are outside of the scope of this work. It is assumed that an on-board CV algorithm is generating relative pose estimates. A transformation has then been performed to convert these estimates into the RSO's body fixed reference frame, and model these quaternion "measurements" as

$$\mathbf{z}_{cv,k} \propto \begin{bmatrix} \frac{1}{2} \mathbf{v}_{cv,k} \\ 1 \end{bmatrix} \otimes (\mathbf{q}_{N'}^B)_k \quad (27)$$

where $\mathbf{v}_{cv,k}$ is small rotation vector to capture the measurement noise. It is taken from a zero mean Gaussian random distribution with covariance

$$\Sigma_{cv,v} = \sigma_{cv,v}^2 \mathbb{I}_{3 \times 3} \quad (28)$$

Based on the best results from the recent satellite pose estimation challenge [36], the standard deviation of the measurement noise, $\sigma_{cv,v}$, is assumed to be 1 deg. These pose measurements are obtained at a rate of 1 Hz.

5.2 The Extended Kalman Filter. The EKF is a ubiquitous method to estimate the state of nonlinear systems based on a first-order Taylor series expansion to linearize the dynamics and measurement functions about the current estimate of the state. Estimation algorithms such as the EKF are very important in the context of guidance and control because the control solution requires knowledge of the position and velocity of the spacecraft to make decisions. Without the estimation component, the spacecraft has no knowledge of where it is in space and thus would be unable to complete its mission in an autonomous fashion.

The EKF takes the form of a two-step process: (1) propagation of the mean and covariance and (2) updating the current mean and covariance estimates based on measurements. While summarized below, the reader is encouraged to refer to Ref. [37] for further details.

Starting with an initial guess of the mean and covariance (which tracks uncertainty in the state), the mean and covariance are propagated with the dynamics up to the time at which a measurement is received. Once a measurement is received, the measurement is compared with the expected measurement. The expected measurement is the output of the measurement function of the sensor using the propagated estimate of the state as the input. Based on the measurement received from the sensor, and the computed expected measurement, the estimate of the state and the associated covariance is updated using the Kalman gain ($\mathbf{K}_k$), which is the optimal linear gain that minimizes the mean square of the a posteriori state estimation error. The mean and covariance are updated with

$$\mathbf{m}_{x,k}^+ = \mathbf{m}_{x,k}^- + \mathbf{K}_k [\Delta \mathbf{z}_k] \quad (29a)$$

$$\begin{aligned} \mathbf{P}_{xx,k}^+ &= [\mathbf{I} - \mathbf{K}_k \mathbf{H}_{x,k}] \mathbf{P}_{xx,k}^- [\mathbf{I} - \mathbf{K}_k \mathbf{H}_{x,k}]^T \\ &+ \mathbf{K}_k \mathbf{H}_{v,k} \mathbf{P}_{vv,k} \mathbf{H}_{v,k}^T \mathbf{K}_k^T \end{aligned} \quad (29b)$$

where $\mathbf{m}_{x,k}^+$ and $\mathbf{P}_{xx,k}^+$ represent the updated mean and covariances, respectively. Here, $\Delta \mathbf{z}_k = \mathbf{z}_k - \mathbf{m}_{z,k}^-$ which is referred to as the

innovation, and $\mathbf{H}_{x,k}$ is the measurement Jacobian with respect to the state, $\mathbf{H}_{v,k}$ is the measurement Jacobian with respect to the noise, and $\mathbf{P}_{vv,k}$ is the measurement noise covariance which in general can be time-varying. For this study, it is constant, with $\mathbf{P}_{vv,k} = \mathbf{\Sigma}_v$. After the update step is performed, the updated mean and covariance are used as the initial conditions for the propagation phase of the next iteration, and the process continues until all the measurements are utilized. The EKF is used to estimate all eight states for the nonlinear dynamics described in Ref. [24]. The estimated nonlinear state vector will henceforth be written as $\hat{\mathbf{x}} = [\hat{x} \ \hat{y} \ \hat{z} \ \hat{\dot{x}} \ \hat{\dot{y}} \ \hat{\dot{z}} \ \hat{R} \ \hat{R}]^T$.

5.3 The Multiplicative Extended Kalman Filter. The MEKF is a modification to the EKF that is often used to estimate a state vector that contains quaternions [38]. The MEKF is very similar to the EKF, barring some slight modifications. It uses the same two-step propagate and update structure, but keeps track of the attitude states and associated covariances in a slightly different manner. The MEKF uses the minimal three-component parameterization of attitude to represent deviations away from some global reference. It represents the attitude as the quaternion product

$$\bar{\mathbf{q}} = \bar{\mathbf{e}}_q(\mathbf{a}) \otimes \bar{\mathbf{q}}_{\text{ref}} \quad (30)$$

where $\bar{\mathbf{q}}_{\text{ref}}$ is a unit reference quaternion, $\bar{\mathbf{e}}_q$ captures the rotation from the reference attitude to the true attitude, and $\mathbf{a}$ is a three-component parameterization of $\bar{\mathbf{e}}_q$. This three-component parameterization of attitude is constructed such that it is a representation of the attitude error and the reference quaternion is an estimate of the true quaternion. This choice for the reference provides a consistent treatment of attitude error statistics by representing the mean of the attitude with $\hat{\mathbf{q}}$ and the covariance of the attitude error with the covariance of the error, $\mathbf{a}$. For this reason, this minimal three-component representation is often called the *error state*.

For attitude representations, the usual formulation of state estimation error ($\mathbf{e}_x = \mathbf{x} - \mathbf{m}_x$) is not valid because representations of attitude are generally not additive, as they do not exist in Euclidean space. Therefore, a multiplicative definition is applied, with the error states

$$\mathbf{e}_x = [(\delta\omega^B)^T \quad (\delta\theta^B)^T]^T \quad (31a)$$

$$\delta\omega^B = \omega^B - \hat{\omega}^B, \quad \delta\theta^B = 2\text{vec}\{\bar{\mathbf{q}}_N^B \otimes (\hat{\mathbf{q}}_N^B)^{-1}\} \quad (31b)$$

Here, $\text{vec}\{\cdot\}$ represents an operator which takes a quaternion and returns the vector portion of the inputted quaternion. The terms $\hat{\omega}^B$ and $\hat{\mathbf{q}}_N^B$ are the estimated states.

Finally, the modifications to the standard EKF propagate and update stages can be observed. The estimated mean still contains the quaternion in the body frame with respect to the inertia frame first introduced in Eq. (2) and is propagated numerically using rigid body dynamics and Eq. (3). The covariance on the other hand is propagated using the error states, and thus in this case will be a 6×6 matrix rather than a 7×7 matrix. Additionally, the update to the attitude representation can no longer be applied additively. The innovation for the attitude measurements takes the form

$$\Delta \mathbf{z}_k = 2\text{vec}\{\mathbf{z}_k \otimes (\mathbf{m}_{x,k}^-)^{-1}\}$$

and is used to update the estimate of the attitude through

$$\mathbf{m}_{x,k}^+ = \left[\begin{array}{c} \frac{1}{2} \Delta \mathbf{z}_k \\ 1 \end{array} \right] \otimes \mathbf{m}_{x,k}^-$$

A final brute-force renormalization step is performed in order to ensure that the quaternion unit norm constraint is maintained. The additive states are updated using Eq. (29a).

6 Control

As presented in Sec. 4, the optimality analysis for the relative motion is conducted with an assumption of the linearized Clohessy–Wiltshire equations. However, the true motion of the servicer is governed by nonlinear dynamics [24]. Hence, relative measurements must be taken and used to estimate the system states so that necessary corrective actions can be taken to ensure the servicer spacecraft follows the desired reference trajectory. With the initial conditions obtained from the guidance Algorithms 1 and 2, a reference trajectory is obtained. The linearized state dynamics of Eq. (1), costate dynamical equations of Eq. (11), and relationship between costate (λ) and inputs ($\mathbf{u}$) of Eq. (12) derived from the first-order necessary conditions are used to propagate the reference trajectory from initial conditions obtained earlier.

An open-loop control-input history is computed for the reference trajectory, however when only these inputs are applied to the servicer, the resulting trajectory under the influence of nonlinear dynamics will depart from the reference trajectory. Hence, a LQR is implemented for trajectory tracking. The controller monitors the deviation of the system states from their intended trajectory and tries to minimize this error by applying desired corrections as necessary. The performance index for the LQR is

$$J(\mathbf{u}) = \int_0^{t_f} (\mathbf{x}_{\text{err}}^T \mathbf{Q} \mathbf{x}_{\text{err}} + \mathbf{u}_{\text{cl}}^T \mathbf{R} \mathbf{u}_{\text{cl}}) dt \quad (32)$$

where $\mathbf{x}_{\text{err}}$ is the error between the estimated states and the reference trajectory, and $\mathbf{u}_{\text{cl}}$ is the additional control input necessary to ensure that the servicer follows the reference trajectory under the influence of the nonlinear dynamics. A gain matrix $\mathbf{K}$ needs to be computed such that the feedback control law $\mathbf{u} = -\mathbf{K} \mathbf{x}_{\text{err}}$ minimizes the performance index given by Eq. (32), subject to linearized system dynamics of Eq. (1). To infer $\mathbf{K}$, the steady-state algebraic Riccati equation is solved for $\mathbf{S}$ and the gain matrix $\mathbf{K}$ is then computed from $\mathbf{S}$. The final control input to the servicer is a superposition of open-loop control $\mathbf{u}_{\text{ol}}$ and closed-loop control $\mathbf{u}_{\text{cl}}$.

$$\mathbf{u} = \mathbf{u}_{\text{ol}} + \mathbf{u}_{\text{cl}} = -\mathbf{B}^T \lambda - \mathbf{K} \mathbf{x}_{\text{err}} \quad (33)$$

Adhering to the overall theorem of optimality, the value of the controller gain is selected to ensure a highly damped response without any considerable overshoots, as it can result in an unnecessary consumption of the on-board fuel. Closed-loop characteristics of the resulting dynamical system can be altered by modifying the tuning parameters $\mathbf{Q}$ and $\mathbf{R}$ of Eq. (32). If low manufacturing tolerance is required, a high value of $\mathbf{Q}$ is needed. This ensures that the servicer follows the reference trajectory closely, albeit with a high consumption of fuel. On the contrary, a high value of $\mathbf{R}$ penalizes the frequent and large thruster firings, conserving the on-board fuel. Generally, a balance between the two objectives is achieved by picking a system response which is sufficiently fast but also does not result into overshoots.

7 Launch and Impact Events

The following section discusses how the discrete events of module launch and impact are modeled and their uncertainties are handled in the simulation of this servicing operation. These events have significant consequences on the relative orbital motion, and must be included in analysis for accurate control performance results. Because of the time scale and velocity difference between orbital motion and module time aloft, modules are assumed to follow straight line trajectories in the relative frame between launch and impact. It is assumed that the mass m_c and inertia tensor $\mathbf{I}^B$ of the RSO, mass of the servicer m_s , and mass of the module m_m are all known to start, though, again, this operational method is conducive to adaptively learning these parameters as well.

7.1 Module Launch. After the servicer has maneuvered to its launch location, a module is then deployed by a mechanism whose

actuation is coincident with the servicer's center of mass. Using conservation of linear momentum, the velocity change the servicer experiences at launch is

$$\mathbf{v}_s^{\mathcal{L}+} = \frac{1}{m_s - m_m} (m_s \mathbf{v}_s^{\mathcal{L}-} - m_m \mathbf{v}_m^{\mathcal{L}}) \quad (34)$$

where $\mathbf{v}_m^{\mathcal{L}}$ is the velocity of the module in the RSO's LVLH frame. If this event were deterministic according to the information used by the guidance algorithm, this velocity would be

$$\mathbf{v}_m^{\mathcal{L}} = \mathbf{R}_B^{\mathcal{L}}(t_i) \cdot (-v_i \hat{\mathbf{n}}_i^{\mathcal{B}}) \quad (35)$$

However, acknowledging velocity uncertainty due to uncertainty in the relative pose measurements taken by the servicer, this process is modeled with an additive zero mean, white velocity noise ν_v such that

$$\tilde{\mathbf{v}}_m^{\mathcal{L}} = \mathbf{R}_B^{\mathcal{L}}(t_i) \cdot (-v_i \hat{\mathbf{n}}_i^{\mathcal{B}}) + \nu_v \quad (36)$$

7.2 Module Impact. Following the module time aloft during which the servicer spacecraft follows an uncontrolled coast arc, the module impacts the RSO. Despite not knowing the target's true attitude with certainty, the module impact time is still assumed to be deterministic for the purpose of this article, i.e., the module's terminal guidance system forces this to be the case. Module impact is modeled as a discrete, perfectly inelastic collision such that the module adheres to the RSO instantaneously upon contact. From conservation of linear momentum, this event imparts a ΔV onto the RSO, changing its inertial orbit. These changes were found to be negligible from previous work [16] and are thus neglected. However, this ΔV when viewed in the relative motion frame becomes significant. While it is the RSO taking the direct state change, the servicer sees an equal and opposite velocity perturbation when the motion is viewed in the RSO's LVLH frame. Hence, the ΔV can be effectively applied to the servicer's relative state while remaining in the same frame. The posterior velocity after impact seen by the servicer is

$$\mathbf{v}_s^{\mathcal{L}+} = \mathbf{v}_s^{\mathcal{L}-} + \frac{m_m}{m_m + m_c} \tilde{\mathbf{v}}_m^{\mathcal{L}} \quad (37)$$

where again, the velocity process noise is used to capture uncertainty.

The module impact also affects the RSO's angular velocity and inertia tensor. These quantities are directly influenced by the impact location in the RSO's body frame and its associated uncertainty. Given an additive zero mean, white uncertainty model, the stochastic impact site is then

$$\tilde{\mathbf{r}}_i^{\mathcal{B}} = \mathbf{r}_i^{\mathcal{B}} + \nu_r \quad (38)$$

The modeled change in the RSO's inertia tensor uses this stochastic quantity in the parallel axis theorem

$$\mathbf{I}^{\mathcal{B}+} = \mathbf{I}^{\mathcal{B}-} - m_m [\tilde{\mathbf{r}}_i^{\mathcal{B}} \times]^2 \quad (39)$$

Finally, using conservation of angular momentum, the posterior angular velocity of the RSO after module impact is

$$\boldsymbol{\omega}^{\mathcal{B}+} = [\mathbf{I}^{\mathcal{B}+}]^{-1} (\mathbf{I}^{\mathcal{B}-} \boldsymbol{\omega}^{\mathcal{B}-} + m_m \tilde{\mathbf{r}}_i^{\mathcal{B}} \times \tilde{\mathbf{v}}_m^{\mathcal{B}}) \quad (40)$$

These quantities are updated in simulation based on estimated states, while true values are updated without process noise based on the truth states. Because of mass exchange, the mass of the RSO and servicer must be book-kept along with the RSO's true inertia tensor.

8 Numerical Simulation

The proposed GN&C suite for the deployment of spacecraft servicing modules is now demonstrated on a tumbling GPS satellite in mid-Earth orbit, where Earth's gravitational parameter is

$\mu = 3.986 \times 10^5 \text{ km}^2 \text{ s}^{-2}$. Only the deployment and impact of modules are examined in this scenario; RSO stabilization and/or inertia parameter adaptation through coordinating module thrust maneuvers are not considered. The simulation is run using the two-body, nonlinear relative equations of motion found in Ref. [24]. Uncertainties in measurements as well as launch and impact event outcomes are included in the simulation and are detailed below.

8.1 Simulation Parameters

8.1.1 Resident Space Object, Servicer, and Module Parameters. The RSO vehicle resides in a slightly eccentric, inclined orbit with semi-major axis $a = 26,938 \text{ km}$, eccentricity $e = 0.075$, inclination $i = 45 \text{ deg}$, right ascension of the ascending node $\Omega = 10 \text{ deg}$, argument of perigee $\nu = 10 \text{ deg}$, and initial true anomaly $f_0 = 0 \text{ deg}$. Its initial mass and inertia tensor, representative of a GPS-sized satellite, are assumed known at $m_c = 2300 \text{ kg}$ and $\mathbf{I}^{\mathcal{B}} = \text{diag}\{[4000, 5000, 6000]\} \text{ kg m}^2$ where the operator $\text{diag}\{\cdot\}$ returns a matrix whose diagonal entries are the elements of the inputted vector. Its initial quaternion and angular velocity are $\tilde{\mathbf{q}}_N^{\mathcal{B}}(t_0) = (1/\sqrt{4})[1, 1, 1, 1]^T$ and $\tilde{\boldsymbol{\omega}}^{\mathcal{B}}(t_0) = [0.05, -0.05, 0.04]^T \text{ rad s}^{-1}$, respectively. The initial quaternion is chosen arbitrarily, and the angular velocity is chosen to be large, representing a spacecraft whose attitude motion is well beyond normal operating limits. For reference, a typical nadir pointing low-Earth orbit satellite has body rates of around 0.002 rad s^{-1} .

The servicer spacecraft has initial mass $m_s = 2000 \text{ kg}$ and is treated as a point mass. Its initial relative orbital state in m and m s^{-1} with respect of the RSO's LVLH frame is

$$\mathbf{x}(t_0) = [-35, 0, 0, 0, 0.01124, 0.01]^T \quad (41)$$

which satisfies the nonlinear relative motion energy matching condition, resulting in a bounded (closed) initial relative orbit [24]. The initial relative distance is chosen small enough such that wide field-of-view cameras and a velocimeter LIDAR can be effectively used in proximity operations after narrow field-of-view cameras were successfully used in the near-rendezvous mission phase prior.

Modules launched from the servicer are known to have a mass $m_m = 5 \text{ kg}$, representative of those developed in Ref. [17] equipped with a full sensor suite and thrust capability. For this simulation, modules are launched from the servicer at a velocity $v_i = 0.5 \text{ m s}^{-1}$ from a distance $r_i = 15 \text{ m}$ away from the desired RSO impact site's surface normal vector. A nominal module launch cadence of $t_c = 4.5834 \text{ h}$ is selected such that four modules are deployed in roughly one RSO orbital period. In searching for optimal relative trajectories, this launch cadence is bound on either side by $t_b = 0.75 \text{ h}$. Paths around the RSO must obey a keep out sphere with radius $r_k = 10 \text{ m}$ centered at the RSO's center of mass. The launch radius is comfortably beyond the keep out radius, giving the guidance algorithm sufficient space to find an appropriate optimal trajectory under the constraints. Depending on an RSO's appendages, and the closed-loop performance of the GN&C system, a larger keep out radius may be required to ensure overall system safety, as reference trajectories tangent to the keep out sphere are never tracked to perfection. For the purposes of demonstration, four modules are deployed onto the RSO during this simulation. The impact sites and corresponding surface normal

Table 1 Module impact sites and surface normal vectors

Module	$\mathbf{r}_i^{\mathcal{B}} \text{ (m)}$	$\hat{\mathbf{n}}_i^{\mathcal{B}}$
1	$[-1.6, -0.8, 0.75]^T$	$[0, 0, 1]^T$
2	$[1.75, 0.75, 0]^T$	$[1, 0, 0]^T$
3	$[-1.5, 1, 0.65]^T$	$[0, 1, 0]^T$
4	$[0, 0, 0.75]^T$	$[0, 0, -1]^T$

vectors in the RSO's body frame are chosen for variety among different faces of the RSO and are displayed in Table 1.

8.1.2 Navigation. At the start of the mission, the components of the initial mean, $\hat{x}_0$, were sampled randomly from various Gaussian distributions. The components of relative position and RSO's orbital radius were sampled around the true initial relative position and orbital radius with a standard deviation of 3.16 m. The relative velocities and radial velocity were sampled around the true initial relative velocity vector and radial velocity with a standard deviation of 0.316 m s^{-1} . The constant power spectral density of the process in Sec. 5.2 was a diagonal matrix with diagonal elements equal to 5×10^{-14} . For the MEKF, the initial estimate of the RSO's attitude was sampled from a normal distribution with standard deviation of $\sqrt{10} \cdot \sigma_{cv,v}$ around the RSO's true attitude. The initial estimate of the RSO's angular velocity was sampled around the true angular velocity vector, with a standard deviation of 0.0316 m s^{-1} . The constant power spectral density of the process in Sec. 5.3 was a diagonal matrix with the first three diagonal elements equal to 5×10^{-25} and the last three diagonal elements equal to 5×10^{-20} .

During the first phase of the mission, underweighting is used to prevent filter divergence and improve stability. Divergent filter behavior refers to cases where the state estimation error is inconsistent with the state uncertainty associated with the estimate. This typically occurs when the estimator receives very precise measurements but has large prior uncertainty in the state estimate. A straightforward method to handle these cases is with the use of underweighting, which was used for the space shuttle navigation system [39]. This underweighting is only performed for a small fraction of the entire mission.

After the guidance algorithm described in Sec. 4 generates the open-loop reference trajectory, the control and navigation algorithms are run simultaneously, guiding the servicer to the launch position. Once the launch occurs, the relative velocity of the servicer will instantaneously be modified. To account for this, the covariance is reset to the initial state covariance used at the start of the mission. After the launch, a coast phase will allow the servicer to continue estimating the relative states while the module is in flight. During this entire period, the MEKF is producing attitude and angular velocity measurements of the RSO. At impact, the angular velocity of the RSO will change instantaneously, and in a similar fashion to the EKF, the covariance will be reset for the MEKF. The converged estimates are then used in the next iteration of the guidance algorithm to generate the new initial conditions for the next launch. This process repeats until all modules are launched.

8.1.3 Control. During the closed-loop maneuver to the launch location, the total control input computed from Eq. (33) is fed to the relative motion EKF. The navigation and control architectures work in tandem to remain close to the trajectory and provide accurate estimates of the relative orbital states of the servicer. During the simulation scenario, the tuning parameters Q and R are, respectively, $\text{diag}\{[.01, .01, .01, 1, 1, 1]\}$ and $\text{diag}\{[100, 100, 100]\}$. The closed-loop poles for the position response are then placed at $-0.087 \pm 0.05j$, resulting in the damping ratio of 0.87 and the damped natural frequency of 0.1 rad s^{-1} for all three coordinates. The controller uses thresholding logic based on the position covariance associated with the state estimate from the EKF to decide whether or not to modify the open-loop control. To account for EKF initialization, the controller will not modify the desired open-loop control signal unless the uncertainty in the position information shrinks below 3.33 m^2 .

8.1.4 Launch and Impact Events. The discrete process noise associated with impact site uncertainty ν_r is modeled to have a diagonal covariance matrix with uniform standard deviations of 10 cm. The discrete process noise associated with velocity change uncertainties ν_v for both launch and impact is modeled to have a diagonal covariance matrix with uniform standard deviations of 0.01 cm^{-1} . These values are based on performance achieved by module relative navigation algorithms in Ref. [18].

8.2 Simulation Results. The resulting closed-loop trajectory followed by the servicer to deploy all four modules is displayed in Fig. 6 in the RSO's LVLH frame, with resulting control shown in Fig. 7. The trajectory plot shows both the estimated and true trajectories. It can be concluded from the figure that both the navigation and control subsystems have acceptable performance that result in tracking errors on the order of cm. The initial spike in tracking error is due to the absence of closed-loop control via a covariance dependent thresholding scheme. In other words, the relative position uncertainties must dip below the predefined value of 3.33 m before the navigation solution can be used for feedback control. During this time, only open-loop control is employed. A result of suddenly including feedback control is seen in the oscillatory behavior of both the truth and estimate early on in the first launch maneuver in Fig. 6.

Figure 6 also shows the geometric consequences of the launch and impact events as they perturb the servicer's relative velocity away from the RSO. The open and closed-loop control must compensate for this dispersion both in planning and execution, respectively. Interestingly, the guidance solution favors mostly in-plane launch locations near $y = 0$ despite the target's attitude motion processing widely in three dimensions. This is likely carry over from the first launch location: if the first location is found to be in a particular plane, successive locations are likely to remain in that plane if possible so that extra control is not used on out-of-plane maneuvering. Lastly, Fig. 6 shows explicitly how a coast arc after launch appears in the operation after the fourth module is deployed.

Figure 8 plots the state estimation errors and associated uncertainties for the relative orbital states generated by the EKF. It can be observed that the filter produces estimates for the position states that are on the order of sub-centimeter level accuracy, which is adequate for the manufacturing tolerances considered here. While the precision of the range measurements are on the order of cm, through measurement accumulation, the position estimates can be made more precise. Up to a point, the longer the filter is able to process measurements, the more precise the estimates will be. In this case, the manufacturing tolerance was less than 1 cm, motivating the maneuver time of over 5 h. Within 5 h, the EKF is able to converge on position estimates that satisfy the desired manufacturing tolerances. For manufacturing operations with even higher tolerances, the filter should be allowed to reach steady state before a launch event is initiated.

The velocity state estimates are also acceptably low, on the order of $1 \times 10^{-5} \text{ m s}^{-1}$. The state estimation errors largely stay within the 3σ covariance intervals, which is a good indicator of a well-behaved sequential estimator. The launch events are also plotted with a white circle, indicating times at which a module was launched. As mentioned above, during these events, the covariance

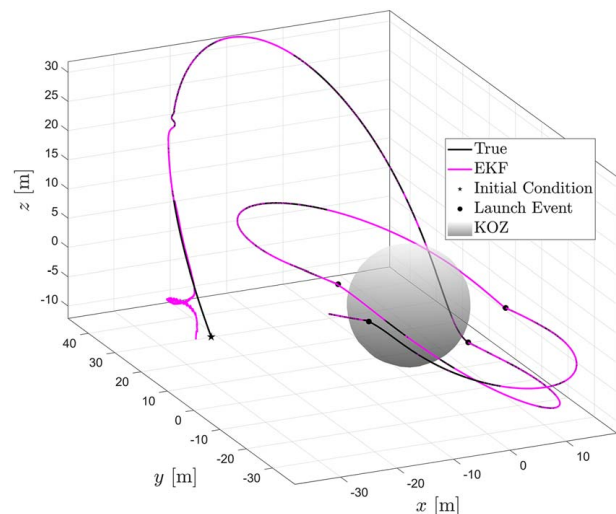


Fig. 6 Configuration space results for full simulation

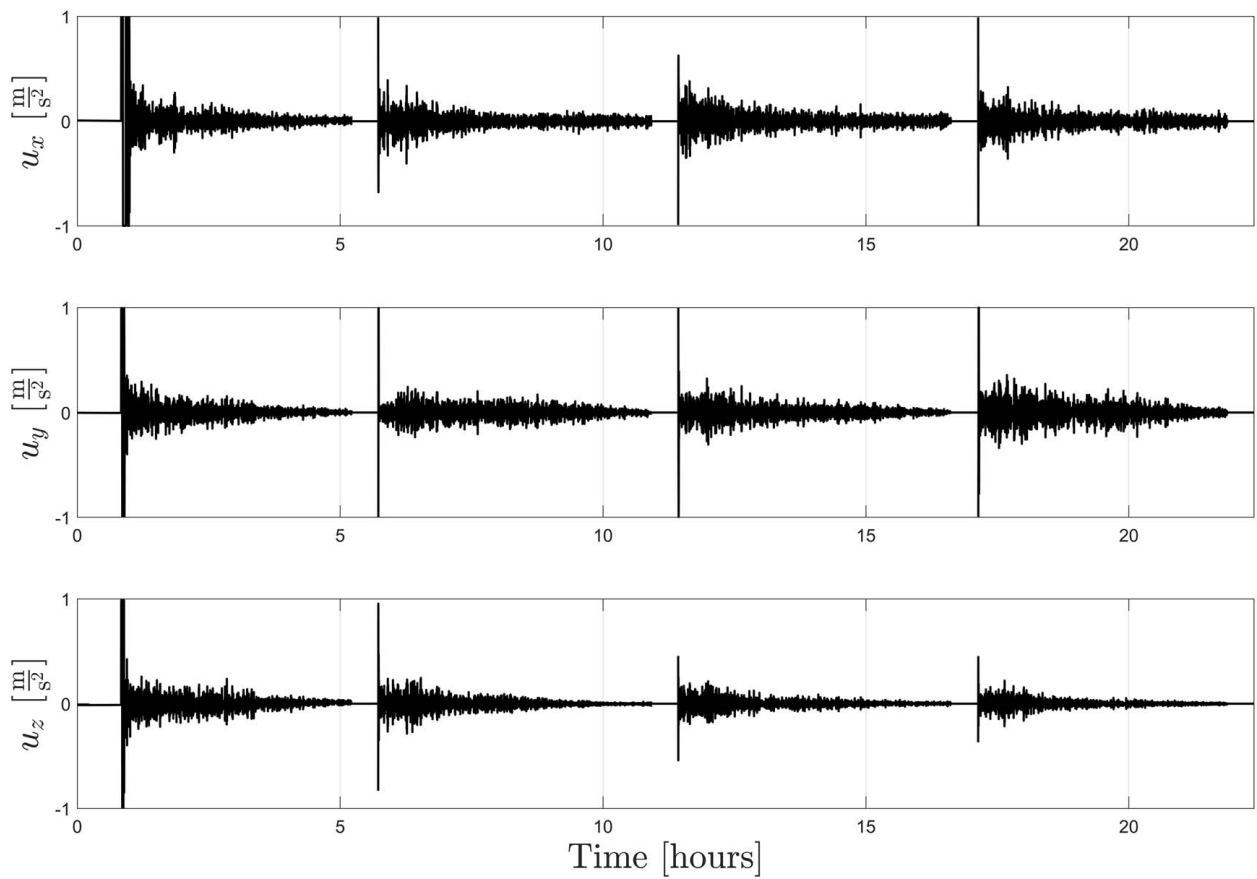


Fig. 7 Control history during full simulation

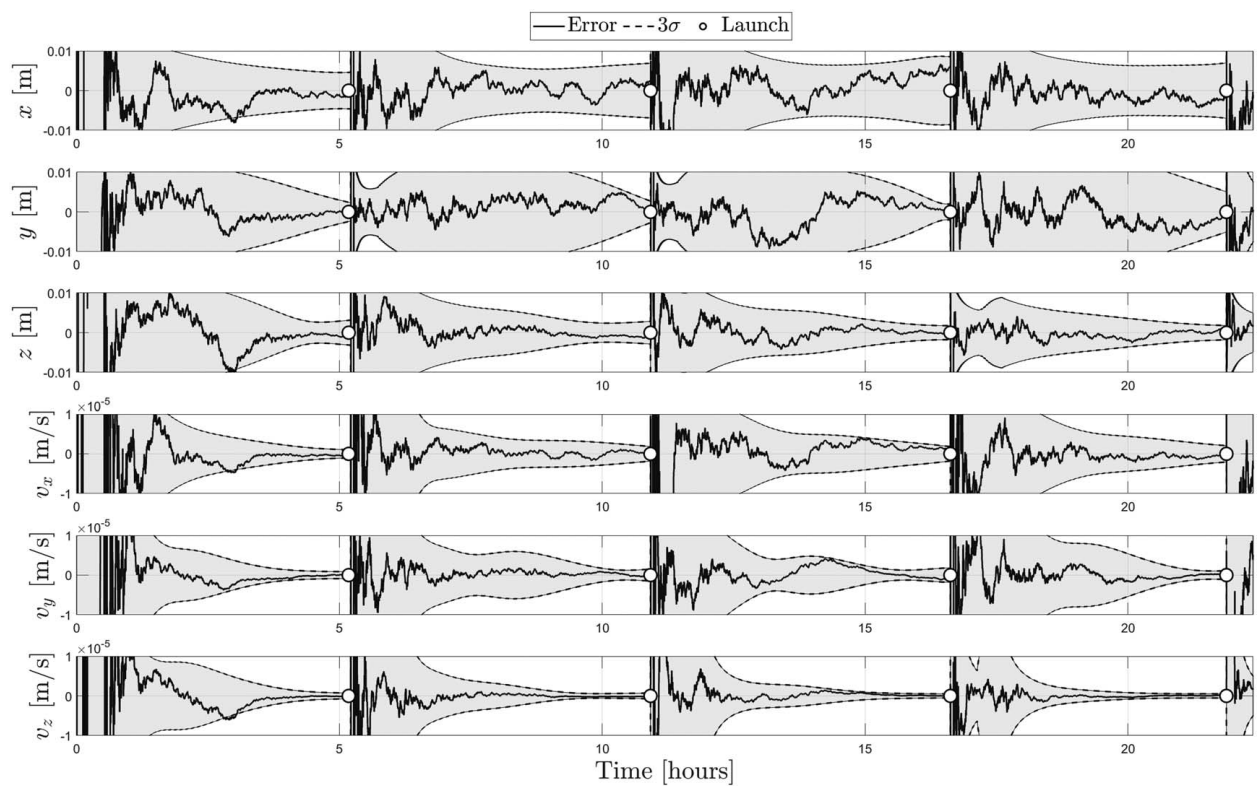


Fig. 8 Relative orbital error states and covariances from the extended Kalman filter

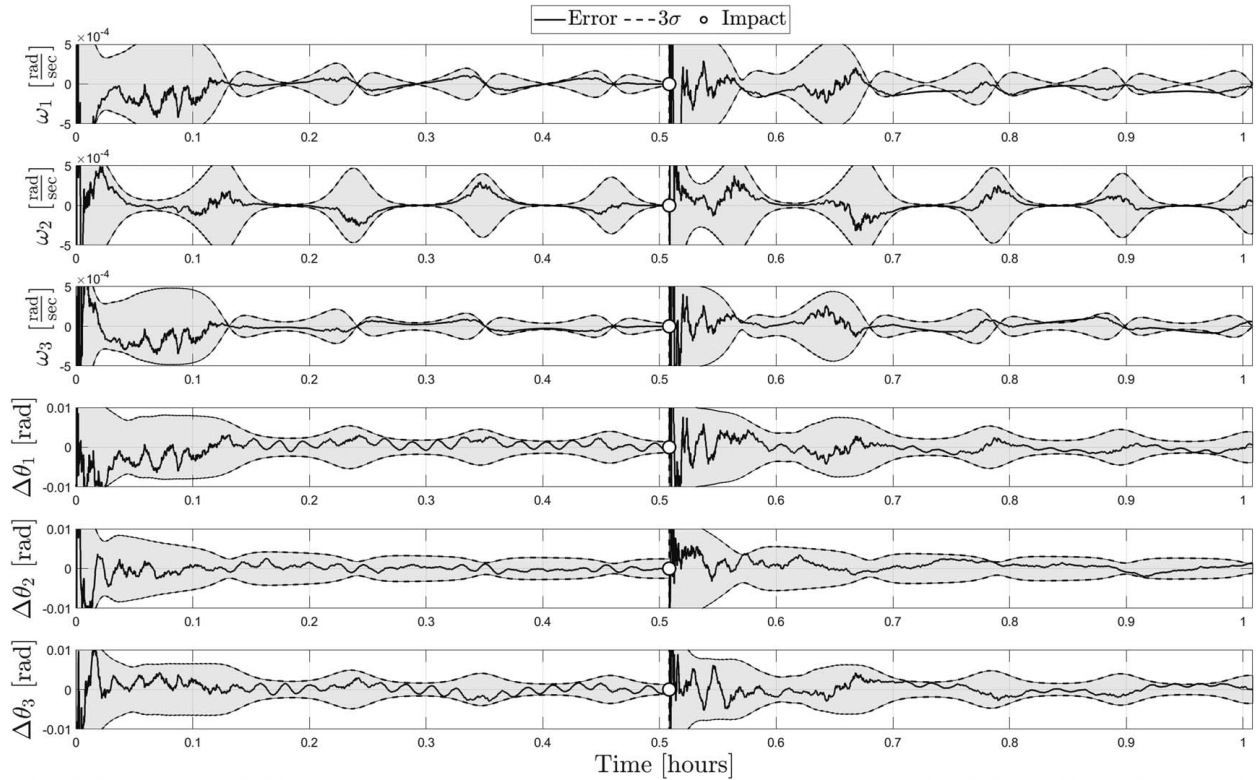


Fig. 9 Attitude error states and covariances from the multiplicative extended Kalman filter during a single coasting and observation phase

is reset to allow the filter to converge on new state estimates, since the velocity states have been altered instantaneously.

Figure 9 plots the state estimation errors and the 3σ of state covariance for the RSO's angular velocity vector and attitude vector, over a single segment of the coast and observation phase described in Sec. 8.1.2. This represents a small segment of the entire mission scenario in which the MEKF generates state estimates for 30 min before and after the impact occurs. The impact time is indicated with a white circle. At this time, the covariance is reset to account for the abrupt change in the angular velocity states that occurs to the RSO. As with the EKF, the state estimation errors largely stay within the 3σ covariance intervals, which is a good indicator of a well-behaved sequential estimator. Additionally, the state estimation errors are relatively low, providing accurate estimates of the RSO's attitude and angular velocity vector, which must be used to generate an open-loop guidance solution for the next launch event.

The control history for the simulation scenario presented in Fig. 6 is shown in Fig. 7. As mentioned above, feedback control is not initially used in order to avoid reacting to any transients in the initial estimation process. Once the position fix is achieved with high accuracy, the closed-loop control kicks in while saturating to the threshold of 10^{-4} m s^{-2} due to large accumulated position error. However, after a few duty cycles, the demanded control forces remain below the allowed maximum and eventually settles to small continuous corrections. At the launch locations, the control system sees an initial saturation spike due to the instantaneous change in velocity imparted by the launch. After this initial spike, the control system makes slight adjustments to follow the desired trajectory. This process repeats for each launch event.

9 Conclusion

This article examined the guidance, navigation, and control required to deploy satellite servicing modules onto a bundle of uncontrolled materials in space as a means of gaining custody as

a precursor to manufacturing/assembly. This bundle was modeled generally as a RSO. A guidance solution that leveraged the computational efficiency of the relative linear dynamics solved for energy optimal trajectories that met both mission time and path constraints. In the numerical simulation, all decision-making was based on *estimated* states of the servicer and the RSO, and an EKF and MEKF were used to generate these estimates based on noisy measurements. Uncertainty in the states and error brought on by nonlinear dynamics were effectively mitigated using an LQR trajectory tracking controller as the servicer spacecraft maneuvered to launch positions.

Module launches and impacts were modeled as discrete events which affected both the relative orbital velocity and the RSO's angular velocity, and had associated launch velocity uncertainties and impact location uncertainties. The launch and impacts were handled in the state estimators by resetting the covariance, and then coasting while state errors converged to near-zero. The simulation results showed successful implementation of the concept of operations, with realistic control accelerations required to complete the operation. This work focused on the high-level implementation of this custody method. Future work should include sufficient target illuminance as a constraint such that both the module and the servicer can observe the RSO with detail in the time surrounding the launch and impact events. Furthermore, the dynamics and control related to module impact and attachment remain an open and important question, and require sufficient exploration before this method can be considered viable. Fully coupled attitude-orbit dynamical models for both the RSO and servicer should be used for higher fidelity results. Overall, this distributed ISAM system architecture presents greater potential for operational flexibility and efficiency when compared to more traditional methods relying only on robotic manipulators fixed to monolithic spacecraft.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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